A2 Chemistry

12[b] Acyl chlorides

Contents:

Candidates should be able to:

Recall the reactions by which acyl chloride can be produced.

Describe reactions of acyl chloride with water, alcohol, phenol, ammonia and amine.

Describe the addition-elimination mechanism of acyl chlorides in reactions.

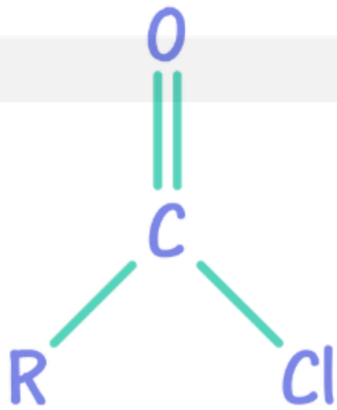
Explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes (Aryl chlorides).

Definitions

Acyl Chloride	organic compounds with a chlorine atom bound to an acyl group

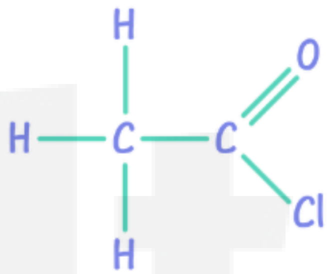
12.1 Acyl Chlorides

The basic structure of an acyl chloride is:

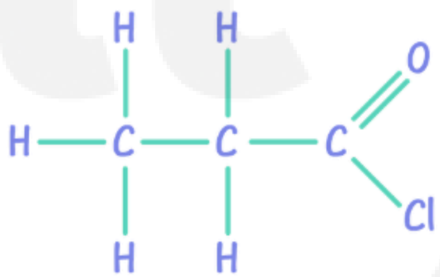


Some examples of compounds are:

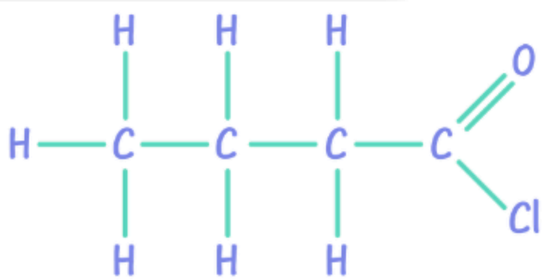
ethanoyl chloride (C_2H_3ClO)
CH_3COCl



propanoyl chloride (C_3H_5ClO)
CH_3CH_2COCl



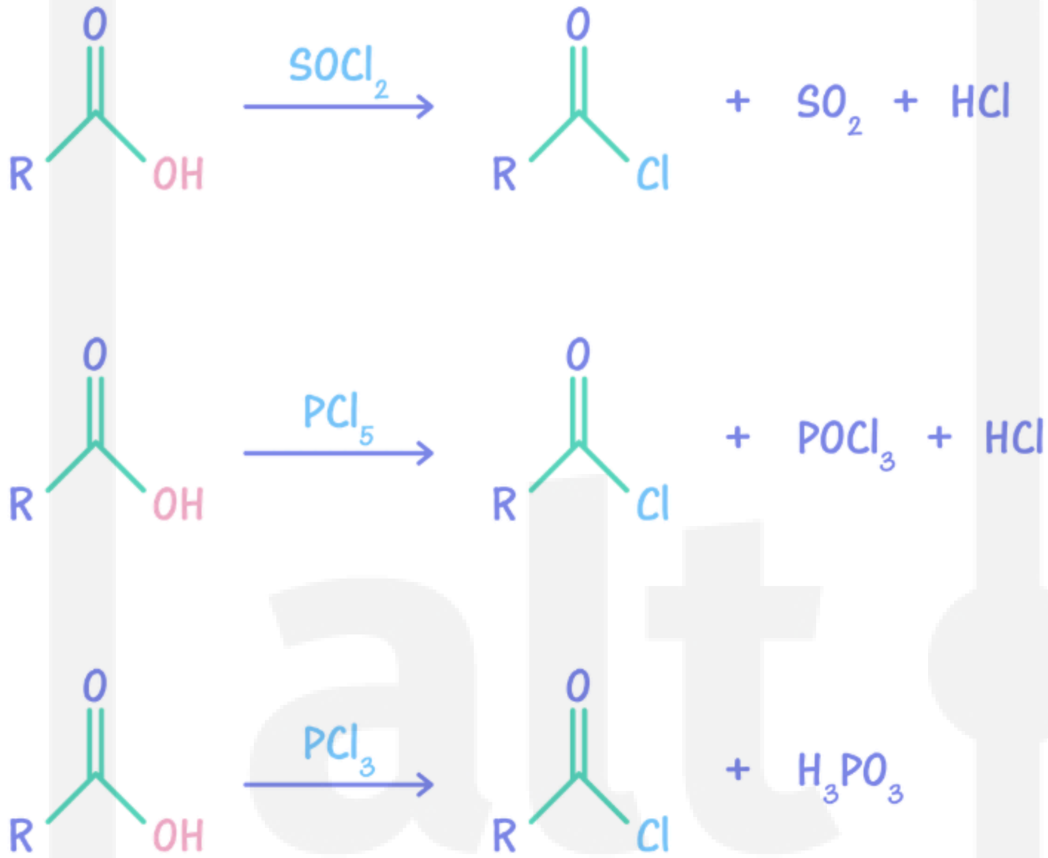
butanoyl chloride (C_4H_7ClO)
$CH_3CH_2CH_2COCl$



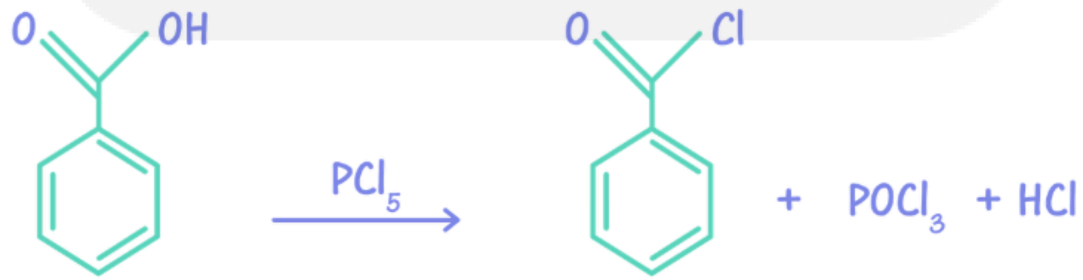
12.1.1 Formation Of Acyl Chlorides:

Phosphorus pentachloride, PCl_5, reacts vigorously with carboxylic acids at room temperature to form acyl chlorides. The -OH group is replaced by a chlorine atom, forming an acyl chloride. The other product of this reaction is HCl gas.

Acyl chlorides can also be formed by heating carboxylic acid with phosphorus trichloride, PCl_3 ,or with sulfur dichloride oxide, $SOCl_2$.



Taking benzoic acid as an example:

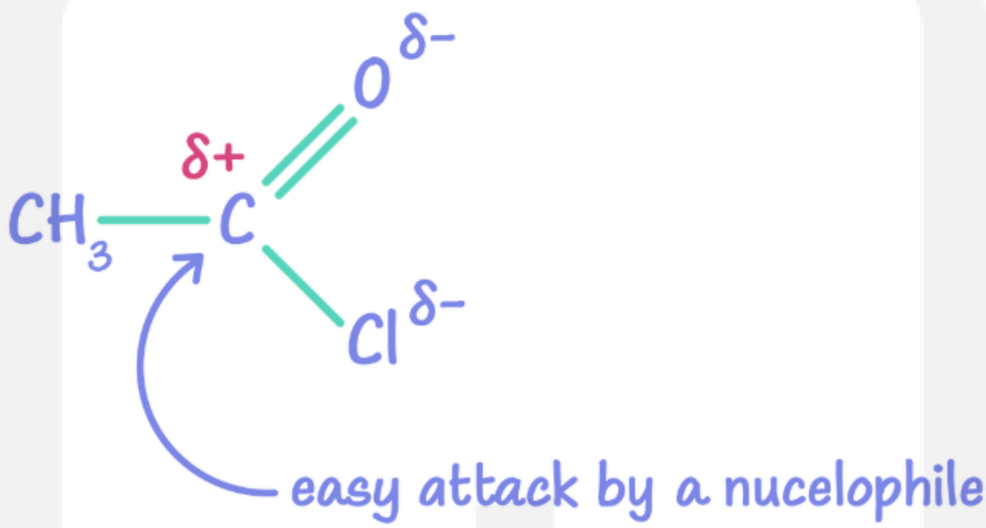


Benzoic Acid

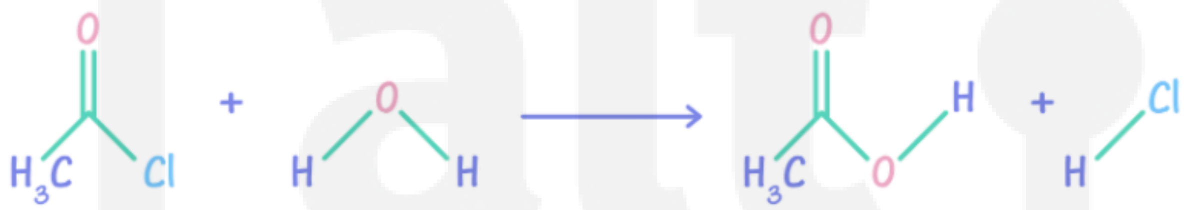
Benzoyl Chloride

12.2 Reactivity

Acyl chlorides are reactive compounds. The electrons on carbonyl carbon are drawn away from it due to strong electronegative effect of chlorine and oxygen atoms. This induces a relatively large partial positive charge on the carbonyl carbon and makes it particularly open to attack from nucleophiles.



Acyl chlorides are hydrolysed by cold water.

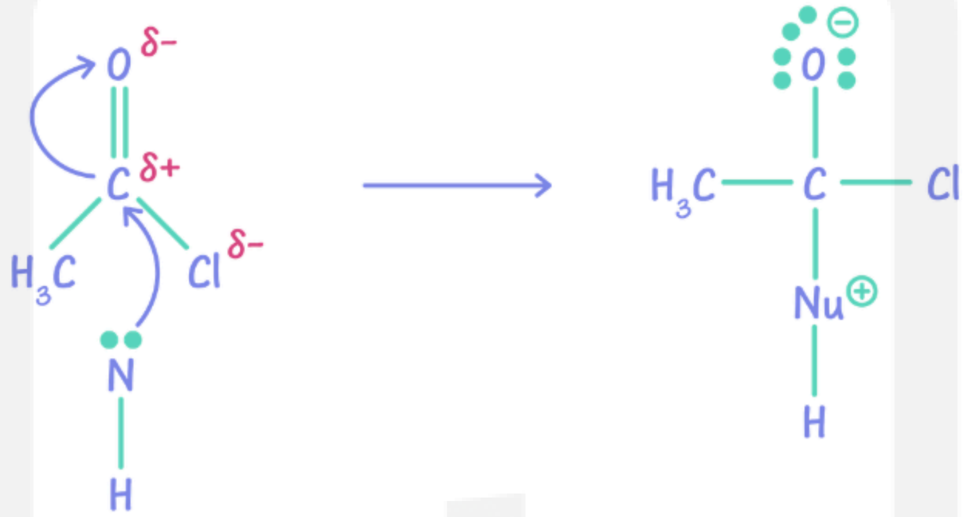


The acyl chlorides react readily with water unlike halogenalkanes and halogenobenzene. This is because the electronegativity of the oxygen atom and the easily polarised C=O double bond in acyl chloride have a dramatic effect on its reactivity. The relative rate of hydrolysis is, **Acyl Chlorides > Halogenalkanes > Halogenobenzene.**

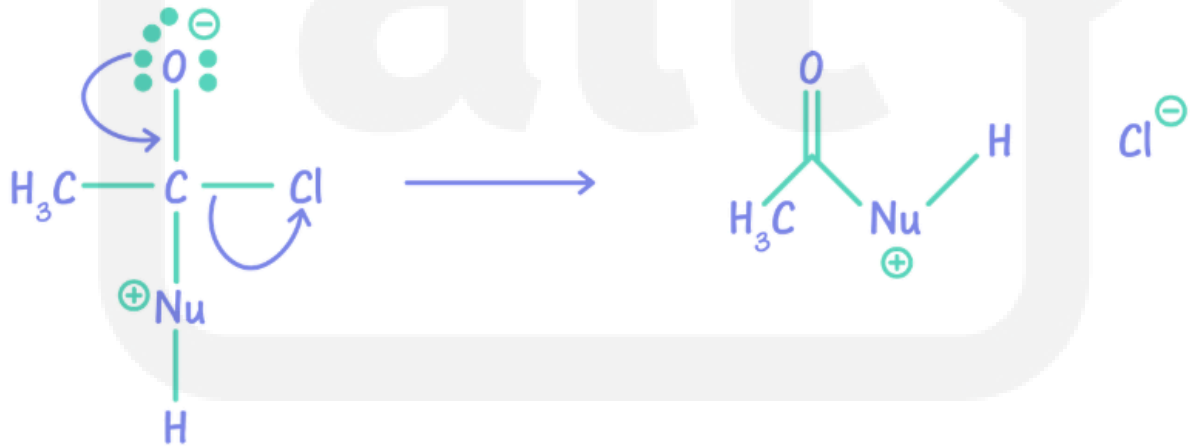
The hydrolysis of acyl chlorides is much faster with alkalis as OH^- is a stronger nucleophile than water.

12.3 Addition Elimination Reaction

The carbonyl group in acyl chlorides can undergo nucleophilic addition in a similar way to carbonyl compounds. In the first step of the reaction, the nucleophile attacks the δ+ C of the C=O group. The π bond breaks with the pair of electrons going to the O to generate O^-.



In the second step, the C-Cl bond breaks, and a lone pair from the Oxygen atom is then used to reform the double bond.



The mechanism of these reactions involve the initial addition of the nucleophile and the elimination of HCl. If the nucleophile is water, then carboxylic acid is formed.

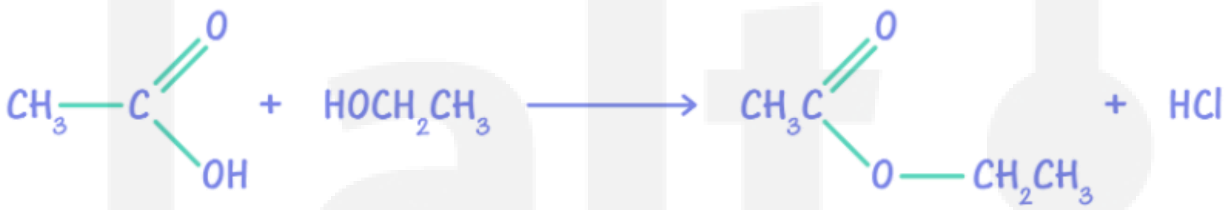
12.3.1 Acyl Chlorides with Alcohols:

Acyl Chlorides react with alcohols to form esters.

Reagent: Alcohol

Condition: rtp

Product: Ester + water

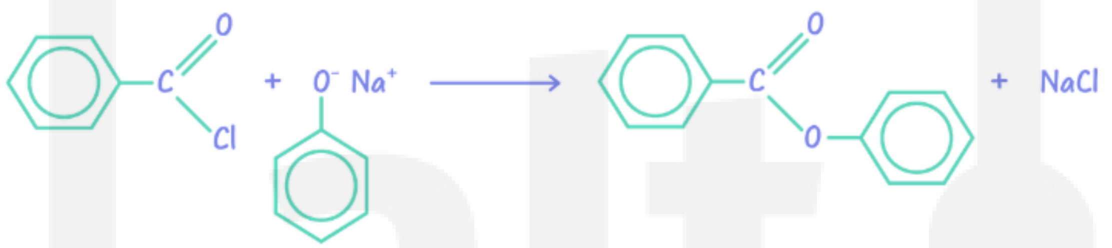


12.3.2 Acyl Chlorides with Phenol:

Alkaline solution of phenol reacts with acyl chloride to form phenyl esters. The phenol is dissolved in NaOH, to make phenoxide which is a stronger nucleophile.

12.3.3 Benzoyl Chloride with Phenol:

Alkaline solution of phenol reacts with benzoyl chloride to form phenyl benzoate. The phenol is dissolved in NaOH, to make phenoxide which is a stronger nucleophile.



12.3.4 Acyl Chlorides with Ammonia:

When concentrated ammonia solution is added to an acyl chloride, a primary amide is formed.

12.3.5 Acyl Chlorides with Amines:

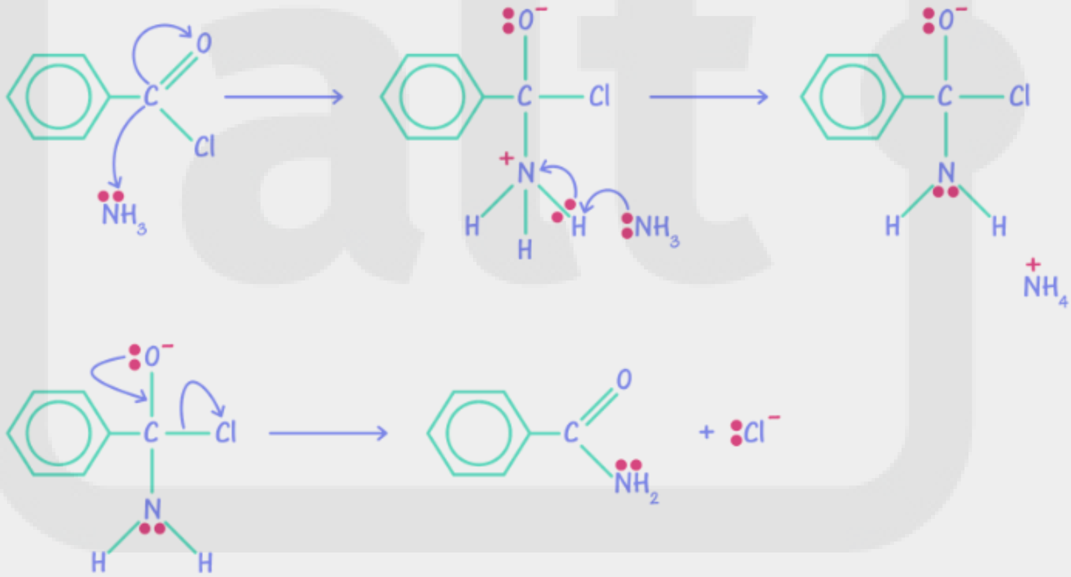
When acyl chlorides react with amines, N-substituted amides are formed.

N-ethylenthanamide

Skill Check 1

Draw the mechanism for the reaction between ammonia and benzoyl chloride:

▼ Solution



Points to Note

- Acyl chlorides are more reactive than halogenalkanes and Halogenobenzenes.
- Acyl chlorides are useful intermediates, forming esters with alcohols or phenols, and amides with amines.